The Learning Ripple Effect of Generative AI

Student Perceptions and Interactive Data Visualization

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ABSTRACT

To explore the impact of conversational chatbots on university students' learning experiences, this project develops an engaging ripple-based visualization. Using Python for data processing, SQLite for data management, and D3.js for visualization, our innovative representations of survey data highlights the dynamic and layered influence of generative AI tools in education.

KEYWORDS

Chatbots; Education; SQLite; Visualization

1. INTRODUCTION

With the rise of chatbots like ChatGPT have gained significant attention for their ability to enhance learning experiences. Their accessibility and versatility make them ideal tools for educational support. However, their limitations, such as potential biases and over-reliance, raise critical questions about their role in higher education. This project explores these dynamics using a structured dataset, aiming to offer actionable insights for educators and students.

2. DATASET DESCRIPTION

The dataset used in this project was obtained from the Kaggle platform, a popular data science and machine learning community. It serves as a hub for collaborative exploration, making it an excellent resource for educational research.

2.1 Dataset Description

The dataset, Chatbots' Impact on University Learning, explores university students' perceptions of generative AI tools like ChatGPT in education.[1] It includes responses from undergraduate and postgraduate students across various academic majors, with a balanced gender distribution, offering a comprehensive view of AI's role in higher education.

The survey uses Likert-scale questions to assess students' knowledge of AI, including its limitations, biases, and emotional intelligence, as well as their willingness to use it for tasks like personalized feedback and improving digital competence. It also addresses concerns such as over-reliance on AI, reduced social interaction, and impacts on transferable skills. Open-ended questions provide qualitative insights into challenges like accuracy and ethical concerns.

Responses are coded numerically (1–5) and grouped into themes like "Digital Competence" and "Concerns" for analysis. Conducted between May 14 and May 31, 2023, the survey collected 131 responses, making it well-suited for ripple-based visualizations that represent the layered effects of students' perceptions and attitudes.

2.2 Reason for Selection

This dataset was chosen for several reasons. First, it focuses on a highly relevant and contemporary topic—generative AI in education—which aligns with the project's objective to explore the influence of AI tools like ChatGPT on university students. The dataset's focus on student perceptions, including both benefits and concerns, provides valuable insights into how these tools are shaping educational practices.

Second, the dataset's structured design, particularly the use of Likert-scale questions, makes it ideal for advanced data visualization techniques. The numerical coding of responses allows for ripple-based visualizations, which can effectively illustrate the intensity and distribution of opinions across different themes. This aligns well with the project's emphasis on creating engaging and meaningful visual representations.

Lastly, the dataset is comprehensive, including a diverse group of respondents from various academic levels and disciplines. This diversity ensures that the findings are representative and applicable to a broad educational context. Additionally, the structured data allows for seamless integration with databases and advanced visualization libraries, enabling the project to deliver interactive and insightful visual outputs.

3. INSPIRATION

Initially, inspired by Valentina D'Efilippo's work "Poppy Field"[2], we explored a flower-shaped visualization, where each petal represented a response level (e.g., 0, 1, 2, 3, 4, 5) and the petal size corresponded to the frequency of that response. However, this approach was not effective due to the narrow range and minimal variation in the data (the characteristics of our data). The small differences in petal sizes resulted in a lack of visual distinction, making it challenging to discern patterns or trends. Moreover, the flower shape did not adequately represent the hierarchical nature of the data to broader themes.

This study illustrates how conversational chatbots create a continuous and expanding impact on learning, much like ripples in water. Therefore, to address the limitations of previous visualization shapes, we adopted a ripple-like design to better visualize the data and represent this dynamic process.

The concentric circles in Figure 1 represent the increasing levels of agreement with students' statements, ranging from "Strongly Disagree" to "Strongly Agree". They symbolize the spreading influence of generative AI in learning, similar to how ripples expand outward when a drop hits water. . Each red dot within the circles represents an individual student's response, and the size of the glowing circle around each dot corresponds to the frequency of chatbot usage, from "Never" to "Very Often". Larger glows indicate higher usage frequency. To provide further insight, hovering over a data point reveals a box at the bottom of the visualization that displays detailed information, including the student's ID, degree, major, and gender.

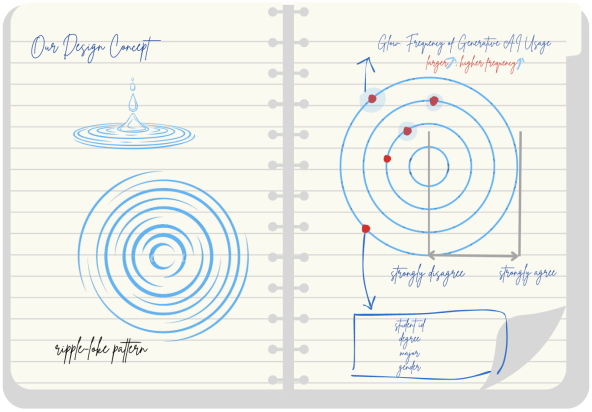


Figure 1. Design Concept

4.TOOLS

4.1 SQLite

SQLite is a lightweight, serverless database engine widely used in data-driven applications. For this project, SQLite was chosen for its simplicity and efficiency in managing small datasets. Its self-contained design eliminates the need for additional server configurations, making it ideal for academic and exploratory projects like this one.

In this project, SQLite served as the primary tool for storing and querying survey data. The database allowed integration with Python3 for preprocessing and D3.js for visualization. With its straightforward implementation and robust SQL support, SQLite enabled efficient data manipulation and ensured a smooth workflow for generating ripple-based visualizations.

4.2 D3.js

D3, the full name is Data-Driven Documents. As one of the powerful libraries of JavaScript, d3.js mainly focuses on data visualization and dynamic graphic display. Using d3.js not only allows developers to create traditional visualizations like histograms, pie charts, line charts, or distribution maps, but it also helps developers utilize graphics such as circles, rectangles, and other shapes to combine and present innovative visual data.

Developers can use the select or selectAll functions to catch the tag name or element on HTML. This is similar to the way of using document.getElementById in traditional DOM. As for selectAll, since it selects all the same tags within the entire HTML document, it is essentially performing a getElement for loop directly through this function. [2]

```
//DOM
var                paragraphs                =
document.getElementByTagName("p")
for(var i=0; i < paragraphs.length; I++){
  var paragraph = paragraphs.item(i);
                  paragraph.style.setProperty(
"color","white", null);
}
```

```
//D3.js
d3.selectAll("p").style("color", "white");
```

This coding style, which uses functions to replace loops, is very efficient, especially when dealing with data, as many datasets often contain thousands or even tens of thousands of entries.

In addition, D3 also has powerful image processing capabilities. In traditional HTML documents, you can use the <svg> tag to draw shapes, such as:

```
<circle cx="60" cy="60" r="10" fill="Red" />
```

cx and cy represent the coordinates of the shape relative to the SVG canvas, where x=60 means 60 units to the right, and y=60 means 60 units down. r represents the radius of the circle. In d3.js, the equivalent way to write this in HTML would be:

```
svg.append("circle")
   .attr("cx", 60)
   .attr("cy", 60)
   .attr("r", 10)
   .style("fill", "Red")
```

This means appending a <circle> tag to the SVG canvas, using the attr method to assign its coordinates and radius, and using the style method to assign CSS styles.

As mentioned in the paragraph above, D3.js is often used in website data visualization. Its advantages in web-based data visualization are obvious, as it integrates HTML, CSS, and JavaScript into a single framework, allowing them to be written together. This also makes it convenient to add other JavaScript interactions or animation functions after creating the visualizations, making D3.js a highly flexible, versatile, and practical tool for rendering data graphics.

5. IMPLEMENTATION

5.1 System Architecture

As mentioned above, due to the advantages of SQLite and d3.js, our project chose to use React.js and Express.js to build the frontend, connecting to APIs to fetch data from the SQLite database. Below is the system architecture diagram for this project.

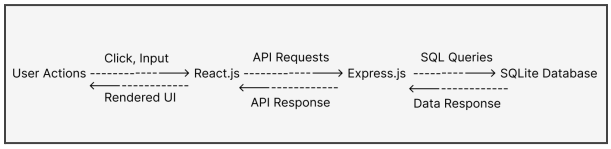


Figure 2. Architecture Diagram

5.2 Data Cleaning & transformation

First, we used Python to read the CSV file and performed data cleaning and transformation on the resulting DataFrame before adding it to our local SQLite database.

During the process, as mentioned in the dataset introduction, this questionnaire data consists of agreement ranks ranging from Strongly Disagree, Disagree, Neutral, Agree, to Strongly Agree. In addition, it includes basic information about each student, such as gender, major, AI chatbot usage frequency, etc.

To visually represent the data by mapping each data point to different radii on concentric circles, we wanted to represent these agreement ranks numerically. Thus, in the Python file, we transformed the data from Strongly Disagree to Strongly Agree into numerical values ranging from 1 to 5, as shown below:

```
agreement_mapping = {
        'Strongly Disagree': 1,
        'Disagree': 2,
        'Neutral': 3,
        'Agree': 4,
        'Strongly Agree': 5
}
```

At the same time, we also transformed the data for students' AI chatbot usage frequency (Q4) into numerical values, as shown below:

```
frequency_mapping = {
        'Never': 1,
        'Rarely': 2,
        'Sometimes': 3,
        'Often': 4,
        'Very Often': 5
}
```

5.3 Data modeling

Since our graphical design represents each node as a student's data point, and each node needs to be positioned based on the degree of agreement for a specific question, the graph rendering in JavaScript requires accessing individual student-related information (such as gender, major, degree, and chatbot usage frequency).

The most convenient way to achieve this is by modifying the data structure. In the modified structure, each question in the questionnaire is placed under a single root as a child, and under each child, the data for each student is stored. This includes the student's name, value, distance, major, gender, degree, and student ID.

```
"name": "Q5.1",
"children": [
{
"name": "Data Point 1",
"value": 4,
"distance": 4.0,
"studentID": "0",
"degree": "Master",
"major":   "International_Economic_Relations",
"gender": "Female"
 }
```

This object-oriented, hierarchical data structure allows for more efficient access to each student's information during the graph rendering process.

5.4 Visualization

In this project, we chose React as the frontend framework and Express.js as the backend framework. The two frameworks exchange modeled data through API requests and responses, making it easier to use the data for visualization with D3.js.

First, we rendered a div tag with the ID chart and used D3.js to select this element. Next, we created an SVG canvas and appended five transparent circle elements as opinion scales.

```
const svg =
d3.select("#chart")
.append("svg")
.attr("width", width)
.attr("height", height);
```

```
svg.append("circle")
.attr("cx", offsetX)
.attr("cy", offsetY)
.attr("r", i * radiusStep)
.style("fill", "none")
.style("stroke", "lightgray") .
style("stroke-dasharray", "4,4")
.style("opacity", 1);
```

Afterward, we used the forEach function to iterate through each piece of children data, drawing the node bodies and the glow effect representing usage frequency within the loop.

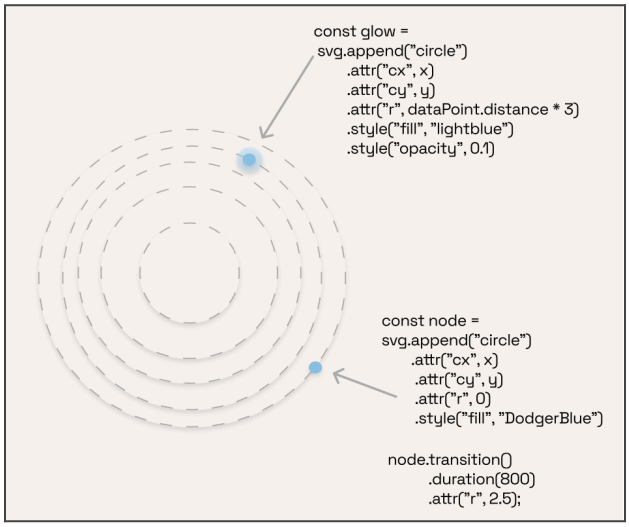


Figure 3. Visualized Graphic

Once the main graphic elements were completed, transitions such as fade-in and fade-out animations could be added to improve user experience and make the visualization more elegant.

6. FUTURE WORK

6.1 Improvements

There are several areas we will improve to enhance the user experience and website readability.

Firstly, the website primarily uses blue as the main color scheme. Certain elements such as the glow around the dots can be

adjusted to contrasting colors like red, which would make readers interact with the interface more effectively.

Secondly, we aim to add keywords to question labels for better clarity. For instance, instead of labeling questions as Q5.1 and Q5.2, we will use descriptive labels like Q5.1 (ChatGPT has limitations in their ability) and Q5.2 (ChatGPT can generate output that is factually inaccurate) to help readers quickly grasp the context.

6.2 Deployment

Apart from the interface design mentioned above, there are also many areas for improvement in the overall website structure. Currently, our website is still running on localhost. If we could further improve it in the future, we hope to deploy the website to an internet URL, making it more convenient for the audience to access.

Regarding the database design, since we are using SQLite, it has the advantages of being lightweight, simple, and highly efficient for small-scale projects. However, it can only serve as a local database. Once the website is deployed to an internet URL (e.g. deployment platforms like Vercel), it will still require a large-scale cloud database such as MongoDB or PostgreSQL.

Therefore, if we aim to further enhance the accessibility of the website, the next step would be to implement database environment configuration.

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REFERENCES

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